

Artificial Intelligence: A Brief Overview

Artificial intelligence (AI) is a branch of computer science that aims to create systems capable of performing tasks that typically require human intelligence. These tasks include learning, reasoning, problem-solving, perception, and natural language understanding.

Machine learning (ML) is a subset of artificial intelligence that focuses on developing algorithms that allow computers to learn from and make predictions based on data. Unlike traditional programming, where rules are explicitly coded, machine learning systems improve their performance through experience and exposure to data.

Deep learning is a specialized form of machine learning that uses artificial neural networks with multiple layers. These deep neural networks are inspired by the structure and function of the human brain. Deep learning has achieved remarkable success in areas such as image recognition, speech processing, and natural language processing.

Natural Language Processing (NLP) is a field at the intersection of computer science, artificial intelligence, and linguistics. NLP focuses on enabling computers to understand, interpret, and generate human language. Applications of NLP include machine translation, sentiment analysis, text summarization, and question answering systems.

Retrieval-Augmented Generation (RAG) is a technique that combines information retrieval with text generation. In a RAG system, relevant documents are first retrieved from a knowledge base, and then a large language model uses the retrieved information as context to generate accurate, grounded answers. RAG helps reduce hallucination by anchoring the model's responses in actual source documents.

The Transformer architecture, introduced in 2017 by Vaswani et al., revolutionized natural language processing. Transformers use a mechanism called self-attention to process input sequences in parallel, rather than sequentially like recurrent neural networks. This architecture forms the foundation of modern large language models such as GPT, BERT, and LLaMA.

Vector embeddings are numerical representations of text in a high-dimensional space. Similar texts are mapped to nearby points in this space, enabling semantic similarity search. Embedding models like Sentence-BERT convert sentences and paragraphs into dense vectors of typically 384 to 768 dimensions. These embeddings are stored in vector databases for efficient retrieval.